

SUPPLY CHAIN ANALYSIS REPORT

Operational Intelligence across Suppliers · Logistics · Quality Control · Revenue

India Region | FY 2024 | 5 Suppliers | 5 Cities

📊 Total Revenue ₹5.78L Across all products	⚠️ Avg Defect Rate 2.28% ▼ 76% avg or poor	🚚 Avg Lead Time 17.1 days Supplier average	📋 Inspection Fail 36 / 100 ▼ 36% fail rate	📦 Total SKUs 100 Unique products
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1. EXECUTIVE SUMMARY

This report presents a comprehensive analysis of the supply chain data for 100 SKUs across three product categories — Skincare, Haircare, and Cosmetics — spanning five major Indian cities. The analysis covers revenue performance, supplier efficiency, logistics costs, quality control, and inventory health.

The supply chain generates a total revenue of ₹5,77,604.82 with an estimated profit of ₹5,19,398.76. However, critical quality and inspection issues require immediate attention — 36% of products failed inspection and 76% of SKUs fall in the Average or Poor quality tier.

Key Highlights

Skincare dominates revenue at ₹2,41,628 (41.8% of total), followed by Haircare at ₹1,74,455 (30.2%) and Cosmetics at ₹1,61,521 (28%). Mumbai and Kolkata are the top-performing cities. Supplier 1 is the best performer with the lowest defect rate of 1.80% and fastest lead time of 14.8 days. Route B is the most expensive transport route at ₹595.66 per unit average.

2. REVENUE ANALYSIS

2.1 Revenue by Product Type

Skincare is the clear revenue leader, contributing nearly 42% of all revenue. Cosmetics, despite having the fewest SKUs (26), shows competitive revenue per unit performance.

Product Type	Total Revenue	% of Total	Avg per SKU
Skincare	₹2,41,628	41.8%	₹6,041
Haircare	₹1,74,455	30.2%	₹5,131
Cosmetics	₹1,61,521	28.0%	₹6,212
TOTAL	₹5,77,605	100%	₹5,776

2.2 Revenue by City Location

Mumbai and Kolkata are nearly tied at the top, together contributing 47.8% of total revenue. Delhi is the weakest market, generating 41% less than Mumbai — a significant regional gap that warrants attention.

Rank	City	Revenue	% of Total	vs Mumbai
#1	Mumbai	₹1,37,755	23.8%	—
#2	Kolkata	₹1,37,078	23.7%	-0.5%
#3	Chennai	₹1,19,143	20.6%	-13.5%
#4	Bangalore	₹1,02,602	17.8%	-25.5%
#5	Delhi	₹81,028	14.0%	-41.2%

3. SUPPLIER PERFORMANCE ANALYSIS

Five suppliers support the entire supply chain. Performance varies significantly in both defect rates and lead times. Supplier 1 is the standout performer while Supplier 5 has the worst defect rate and Supplier 3 has the longest lead time.

Supplier	Revenue	Avg Defect Rate	Avg Lead Time	Rating
Supplier 1	₹1,57,529	1.80%	14.8 days	✓ Best
Supplier 4	₹86,469	2.34%	15.2 days	Average
Supplier 2	₹1,25,467	2.36%	18.5 days	Average
Supplier 3	₹97,796	2.47%	20.1 days	Poor
Supplier 5	₹1,10,343	2.67%	18.1 days	X Worst

Key Supplier Observations

- Supplier 1 is the best on both metrics — lowest defect (1.80%) and fastest delivery (14.8 days). Prioritize orders through Supplier 1.
- Supplier 5 has the worst defect rate (2.67%) despite a short lead time (14.72 days) — speed without quality. Needs quality audit.
- Supplier 3 has the longest lead time at 20.1 days — 36% slower than Supplier 1. Consider reducing dependency on Supplier 3 for time-sensitive products.
- Overall defect average is 2.28% — above the ideal threshold of 1.5% for FMCG supply chains. Systematic quality improvement needed.

4. QUALITY CONTROL & INSPECTION ANALYSIS

4.1 Inspection Results

The inspection results reveal a serious quality problem — only 23% of products passed inspection. 36% failed outright and 41% are still pending, meaning the actual fail rate could be even higher once pending reviews are completed.

Inspection Result	Count	% of Total	Status
Pending	41	41%	⚠ Under Review
Fail	36	36%	✗ Failed
Pass	23	23%	✓ Cleared

4.2 Quality Tier Distribution

Products are classified into three quality tiers based on defect rates. The distribution shows 76% of all SKUs are in the Average or Poor tier — a critical finding that demands urgent action.

Quality Tier	SKU Count	% of Total	Defect Range
Good (< 1% defect)	24	24%	Below 1.0% — Excellent quality
Average (1–3% defect)	44	44%	1.0% to 3.0% — Needs monitoring
Poor (> 3% defect)	32	32%	Above 3.0% — Unacceptable level

Quality Recommendations

- Immediately investigate the 36 failed SKUs — identify common suppliers or product categories causing failures.
- Resolve the 41 pending inspections as priority — the true fail rate may be significantly higher than 36%.
- Set a target defect rate of under 1.5% across all suppliers within 2 quarters and introduce monthly quality audits.

5. LOGISTICS & SHIPPING ANALYSIS

5.1 Transportation Modes

Road transport dominates at 29%, closely followed by Rail (28%) and Air (26%). Sea is the least used at 17%. The near-equal split between Road, Rail, and Air suggests opportunities for cost optimization by shifting volume toward Rail.

Mode	SKU Count	% Share	Cost Implication
Road	29	29%	Flexible, moderate cost, door-to-door
Rail	28	28%	Most cost-efficient for bulk shipments
Air	26	26%	Fastest but most expensive option
Sea	17	17%	Cheapest but slowest — ideal for non-urgent

5.2 Route Cost Analysis

Route B is the most expensive at ₹595.66 per unit — 22.7% more expensive than Route A. This gap is significant at scale and should be reviewed for consolidation or renegotiation.

Route	Avg Cost/Unit	vs Cheapest	Recommendation
Route A	₹485.48	Cheapest	Preferred route — maximize use
Route C	₹500.47	+3.1%	Acceptable — use as backup
Route B	₹595.66	+22.7%	Review & renegotiate urgently

5.3 Shipping Delays

17 out of 100 orders (17%) experienced shipping delays — where shipping time exceeded the customer lead time. This is a significant service level issue. The average shipping cost across all carriers is ₹5.55 with minimal variation between Carrier A (₹5.55), Carrier B (₹5.51), and Carrier C (₹5.60).

6. INVENTORY ANALYSIS

Overall inventory health is moderate. The average stock level is 47.77 units with an average order quantity of 49.22 — suggesting orders are roughly aligned with stock levels. However, product-level differences reveal potential overstocking in Cosmetics and understocking in Skincare.

Product	Avg Stock	Avg Order Qty	Mfg Cost	Status
Skincare	40.2 units	49.22	₹48.99	⚠ Low stock
Haircare	48.4 units	49.22	₹48.46	✓ Balanced
Cosmetics	58.7 units	49.22	₹43.05	↑ Overstocked

Skincare has the lowest average stock (40.2 units) despite being the top revenue product — this is a risk. If demand spikes, stockouts could lead to lost sales. Cosmetics has 46% more stock than Skincare but generates less revenue — indicating potential overstock and tied-up capital.

7. KEY INSIGHTS

The following four insights represent the most critical findings from this supply chain analysis:

INSIGHT 1: Supplier 1 is the clear best performer

With a defect rate of just 1.80% (vs 2.67% average of worst supplier) and the fastest lead time of 14.8 days, Supplier 1 should be the primary supplier for high-value SKUs. Increasing order allocation to Supplier 1 could reduce overall defect rates by an estimated 0.4–0.5 percentage points.

INSIGHT 2: Skincare drives revenue but faces inventory risk

Skincare generates ₹2,41,628 — 42% of total revenue — yet carries the lowest average stock of just 40.2 units across 40 SKUs. This mismatch between revenue contribution and inventory depth is a supply chain vulnerability. A stockout in skincare would have a disproportionate revenue impact compared to other categories.

INSIGHT 3: Quality control is the most urgent problem

Only 23 out of 100 products passed inspection — a 77% non-pass rate. With 36 products failing and 41 still pending, the supply chain has a systemic quality issue. Cosmetics has the best defect rate at 1.92% while Haircare is worst at 2.48%. Immediate root cause analysis and supplier quality audits are required.

INSIGHT 4: Route B is draining logistics budget unnecessarily

Route B costs ₹595.66 per unit on average — 22.7% more expensive than Route A at ₹485.48. With 100 SKUs in the dataset, shifting even 20% of Route B shipments to Route A could save approximately ₹2,200+ per cycle. Renegotiating Route B contracts or consolidating shipments on Route A should be a priority cost-reduction action.

8. RECOMMENDATIONS

#	Area	Recommendation	Priority
1	Supplier Quality	Conduct quality audit on Supplier 5 (2.67% defect rate). Set contractual defect cap of 2.0%.	 High
2	Inspection Backlog	Clear 41 pending inspections immediately. Real fail rate may exceed 36% once resolved.	 High
3	Route Optimization	Shift shipments from Route B to Route A. Potential 22.7% logistics cost saving.	 Medium
4	Skincare Inventory	Increase Skincare stock buffer by 20–25% to protect the top-revenue category from stockouts.	 Medium
5	Supplier Allocation	Increase order allocation to Supplier 1 for high-value SKUs. Best quality and fastest delivery.	 Low
6	Delhi Market	Investigate why Delhi revenue is 41% below Mumbai. Consider targeted promotions or channel review.	 Low